
CERTGUARDIAN: SSL CERTIFICATE EXPIRY WATCHER

Flow chart and Use Case Diagram Report

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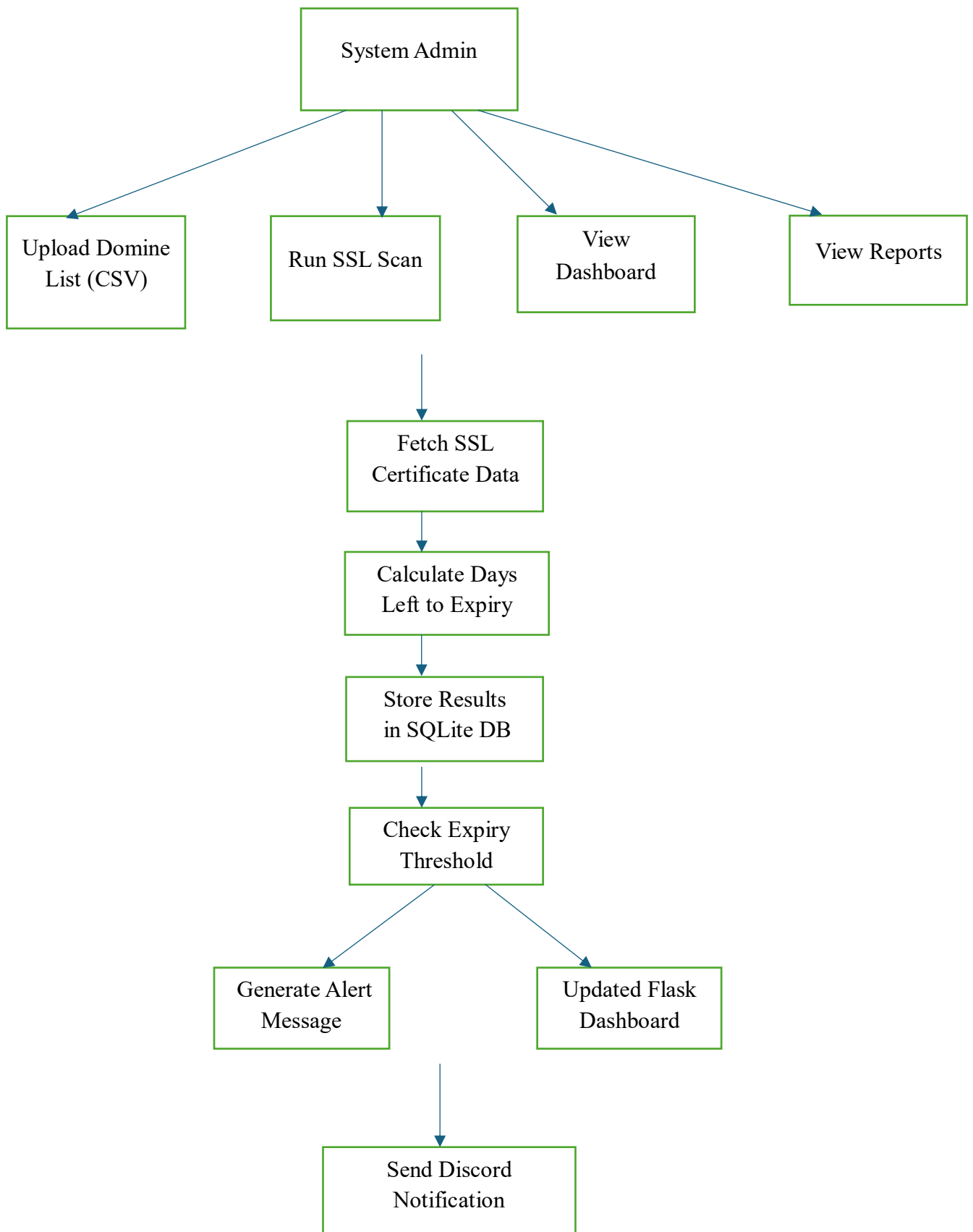
FLOW CHART

The SSL Certificate Expiry Watcher is designed to monitor SSL certificates of multiple websites and notify administrators before certificates expire. The primary actor in the system is the System Administrator, who uploads a list of domains, initiates SSL scans, views monitoring results, and receives alerts.

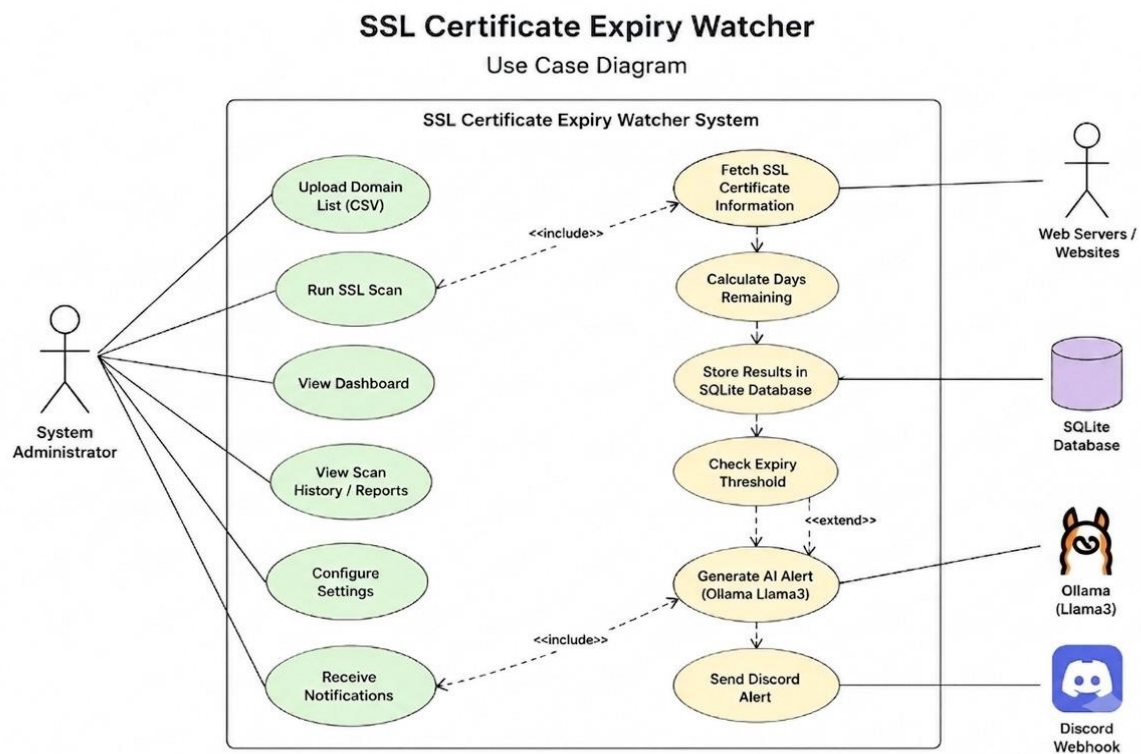
The process begins when the administrator uploads a CSV file containing domain names and runs the SSL scan. The system then connects to each website and fetches SSL certificate information, including the certificate expiry date. After retrieving the certificate details, the system calculates the number of days remaining before the certificate expires.

Once the alert message is generated, the system sends a notification through a Discord webhook so that the administrator can take immediate action. Simultaneously, all monitoring results are displayed on the Flask dashboard, where the administrator can view current SSL status, expiry dates, warning levels, and historical scan reports.

The external entities involved in the system include public websites that provide SSL certificates, the SQLite database used for storing monitoring data, the Ollama Llama3 model used for generating AI-based alerts, and the Discord webhook service used for sending notifications.



Use Case Diagram



Actors

Primary Actor

System Administrator

Uploads domain list

Runs scans

Views dashboard

Receives alerts

Secondary Actors

Public Websites

Provide SSL certificates

SQLite Database

Stores monitoring results

Ollama (Llama3)

Generates alert messages

Discord Webhook

Delivers notifications

Use Case Report

Use Case 1: Upload Domain List

Actor

System Administrator

Description

Administrator uploads a CSV file containing domains to be monitored.

Preconditions

CSV file exists.

Domains are valid.

Postconditions

Domains are available for scanning.

Use Case 2: Run SSL Scan

Actor

System Administrator

Description

System connects to each domain and retrieves SSL certificate information.

Preconditions

Internet connection available.

Domain list loaded.

Postconditions

SSL information retrieved.

Use Case 3: Store SSL Data

Actor

System

Description

Stores SSL expiry details and scan results into SQLite database.

Preconditions

SSL information successfully collected.

Postconditions

Records stored in database.

Use Case 4: Generate Alert

Actor

System

Description

Generates AI-based alert messages for domains nearing certificate expiration.

Preconditions

Days remaining below threshold.

Postconditions

Alert message created.

Use Case 5: Send Notification**Actor**

System

Description

Alert is delivered through Discord webhook.

Preconditions

Alert generated.

Postconditions

Administrator notified.

Use Case 6: View Dashboard**Actor**

System Administrator

Description

Administrator views SSL status, expiry dates, and alerts through Flask dashboard.

Preconditions

Flask application running.

Postconditions

Monitoring information displayed.